

Chapter 1

Adaptive Control of a Real DC Motor

1.1 Introduction

In the previous example of the reader, the adaptive control of a DC-Motor is implemented on an Arduino Due using a first order model. In this experiment the adaptive control of a DC-Motor is implemented on an ESP32 using a second order model. The hardware consists of an ESP32, an L298N H-bridge, and a DC motor with a quadrature encoder. For coding the ESP32, Simulink-Code-Generation is used. Figure 1.1 demonstrates the setup of the experiment. The plant parameters are estimated recursively, the controller is redesigned from these estimates, and the resulting incremental PI algorithm is executed on the ESP32 input/output interface.

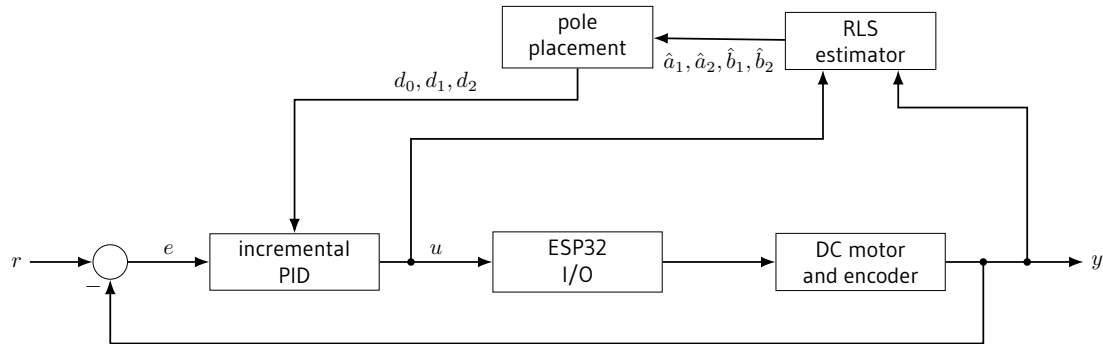


Figure 1.1: Self-tuning regulator used for the ESP32 DC motor rig.

A more comprehensive DC motor model, accounting for electrical dynamics (armature inductance) in addition to mechanical dynamics, may be modeled as a second-order system. Normalizing the equation by dividing by JL_a yields a standard monic form:

$$\ddot{\omega} + \left(\frac{R_a}{L_a} + \frac{b}{J} \right) \dot{\omega} + \left(\frac{bR_a + K_m K_b}{JL_a} \right) \omega = \frac{K_m}{JL_a} u_{in} \quad (1.1)$$

This translates into a general second-order continuous transfer function:

$$G(s) = \frac{K}{(s + p_1)(s + p_2)} \quad (1.2)$$

Discretization yields the discrete transfer function:

$$G(z) = \frac{b_1 z + b_2}{z^2 + a_1 z + a_2} \quad (1.3)$$

If we choose an incremental PID controller to handle the higher-order plant:

$$G_C(z) = \frac{d_2 z^2 + d_1 z + d_0}{z(z - 1)} \quad (1.4)$$

The closed-loop transfer function becomes:

$$T(z) = \frac{(b_1 z + b_2)(d_2 z^2 + d_1 z + d_0)}{z^4 + (a_1 - 1 + b_1 d_2)z^3 + (a_2 - a_1 + b_1 d_1 + b_2 d_2)z^2 + (b_1 d_0 + b_2 d_1 - a_2)z + b_2 d_0} \quad (1.5)$$

For a desired fourth-order denominator polynomial:

$$q(z) = z^4 + q_3 z^3 + q_2 z^2 + q_1 z + q_0 \quad (1.6)$$

We equate the coefficients to derive the Diophantine system of equations. Solving top-down for the dominant controller parameters (d_2, d_1, d_0) yields:

$$d_2 = \frac{q_3 - a_1 + 1}{b_1} \quad (1.7)$$

$$d_1 = \frac{q_2 - a_2 + a_1 - b_2 d_2}{b_1} \quad (1.8)$$

$$d_0 = \frac{q_1 + a_2 - b_2 d_1}{b_1} \quad (1.9)$$

1.2 The Test Rig

The test rig, as can be seen in Figures 1.3 and 1.3, is to be powered by a power supply with standard power jack connector, where the supply voltage must be greater than 12 V, since the DC-Motor can be powered up to 12 V. The voltage from the power supply is converted to 12 V by a buck converter. The ESP32 is to be connected to the notebook via the USB type C connector, where the cable used can not be a USB-C to USB-C cable but rather a USB-A to USB-C, because of the pinout of the connector used in the ESP32 Dev Board.

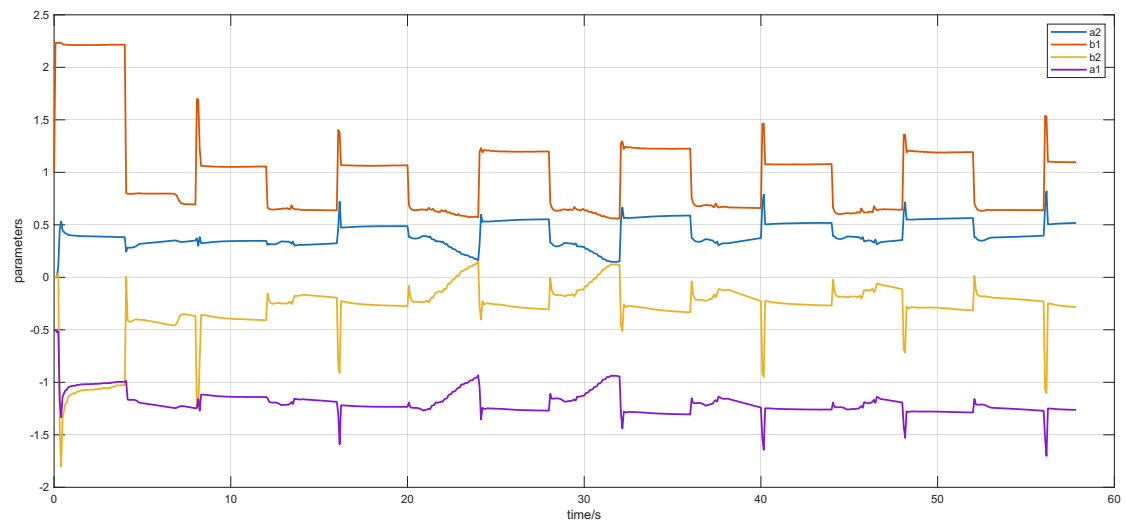


Figure 1.2

**Figure 1.3**

1.3 Measurements

Figure 1.4 shows the convergence of the estimated plant parameters during operation.

**Figure 1.4:** Estimated parameters of the DC motor.

The corresponding closed-loop tracking response of the system is illustrated in Figure 1.5.

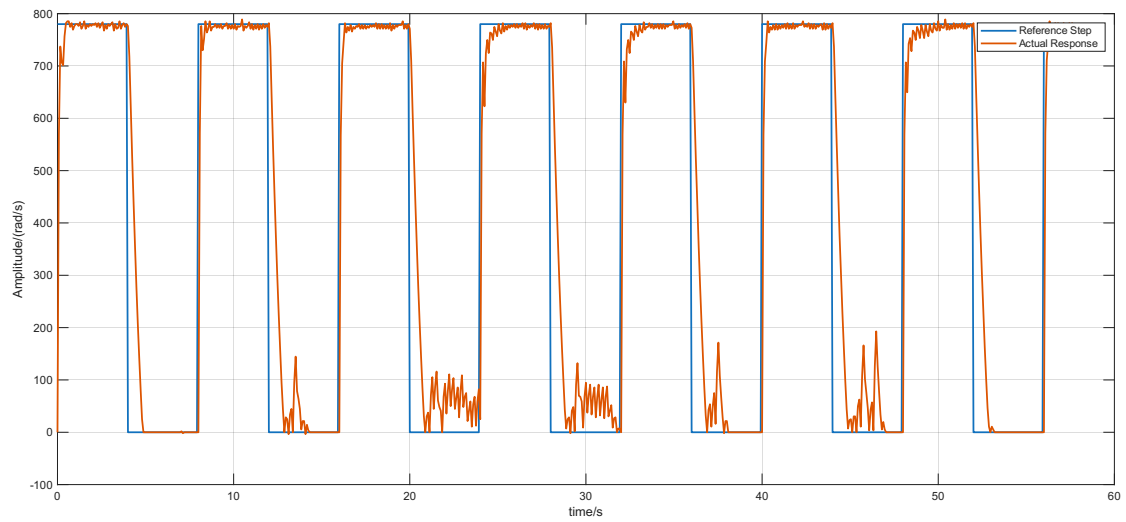


Figure 1.5: Tracking response of the adaptive controller.